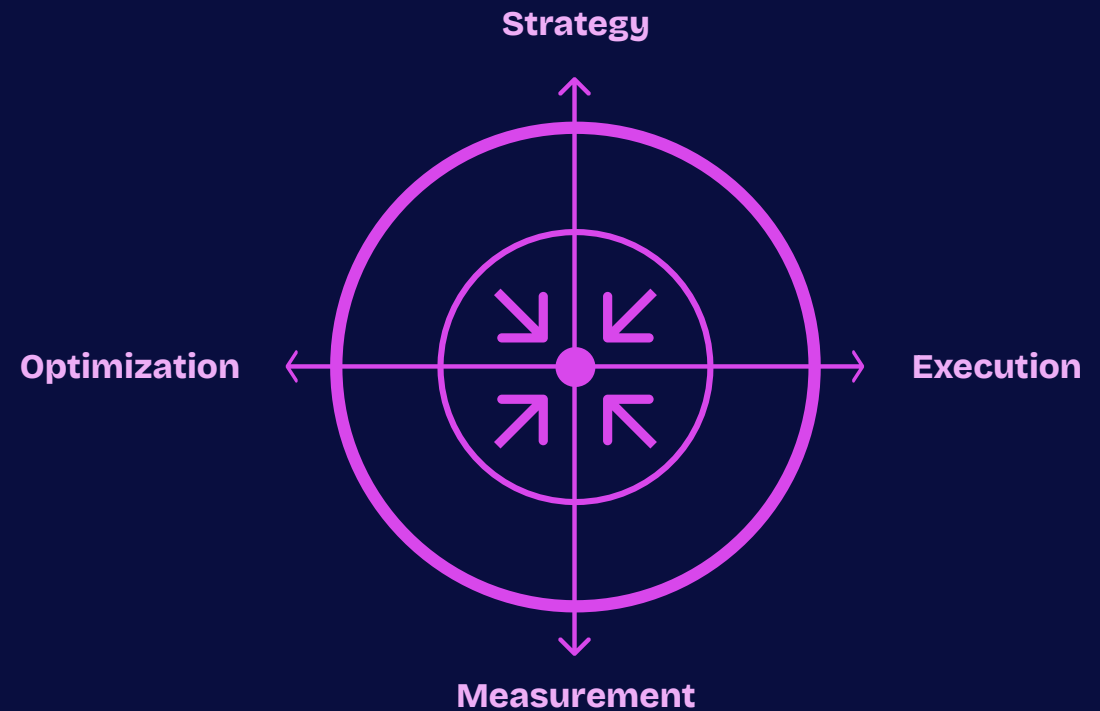


Olist E-Commerce Analytics

SQL → Python → Power BI: A Complete Data Pipeline



The Business Challenge

Scale

100,000+ orders across Brazil's largest online marketplace

Revenue

Where is revenue coming from — and how is it growing?

Customers

Who are our buyers and how loyal are they?

Operations

How reliable is delivery? What actually drives satisfaction?



R\$15.4M in Gross Merchandise Value

96K

Orders Delivered

Sep 2016 – Oct 2018

9x

Revenue Growth

From early 2017 to peak

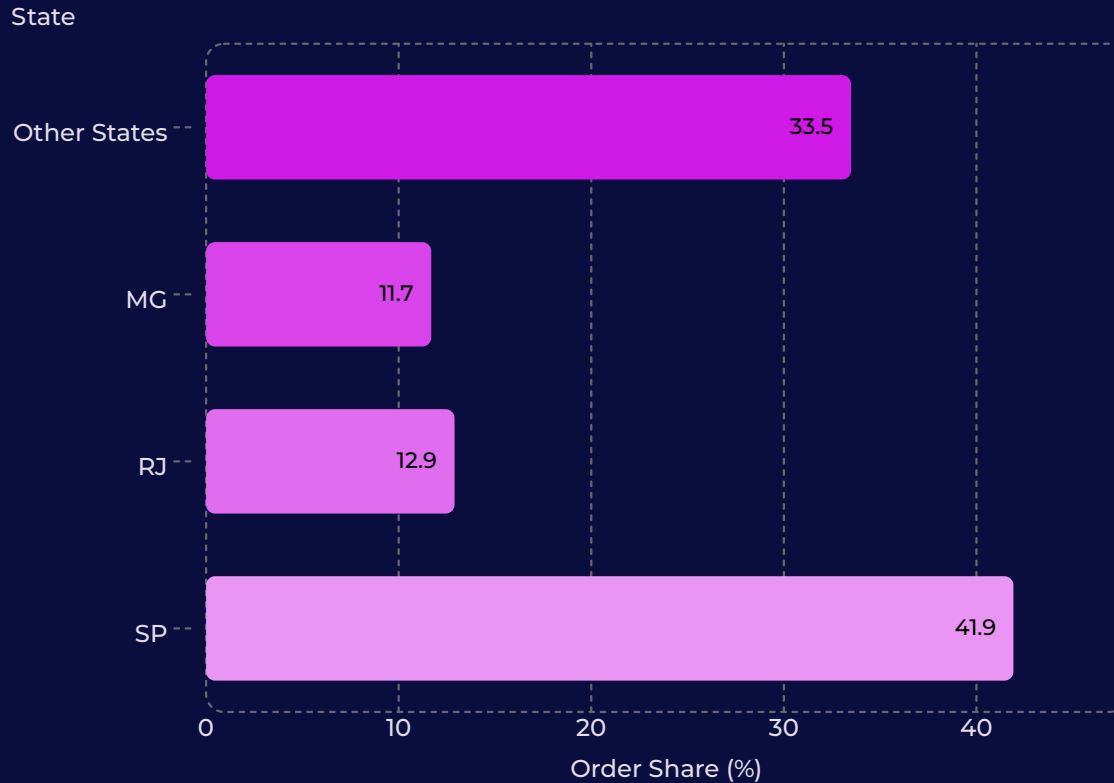
R\$15.4M

Total GMV

Gross Merchandise Value

Revenue scaled aggressively from early 2017, stabilizing into 2018 — with a clear **Black Friday spike in November 2017** confirming strong seasonality in the market.

Geographic Concentration: A Strength & Risk



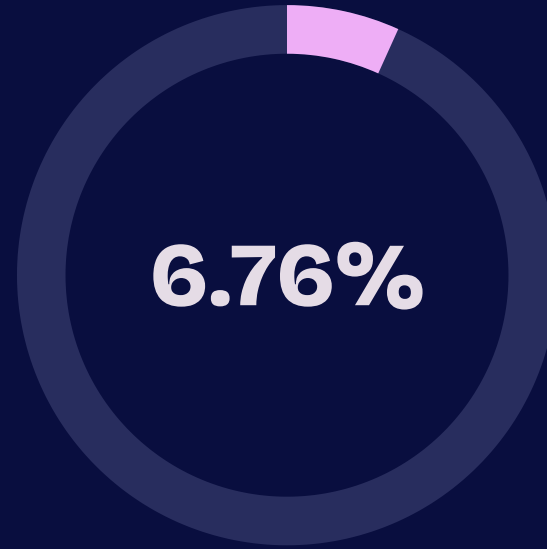
What This Tells Us

São Paulo alone: 41.92% of all orders — a commanding lead that signals both market dominance and seller concentration.

The **top 3 states (SP, RJ, MG)** account for **66.51%** of total volume.

Expansion opportunity is real — but so is the risk of over-reliance on a single region.

Delivery Performance: Geography Matters



Late Delivery Rate

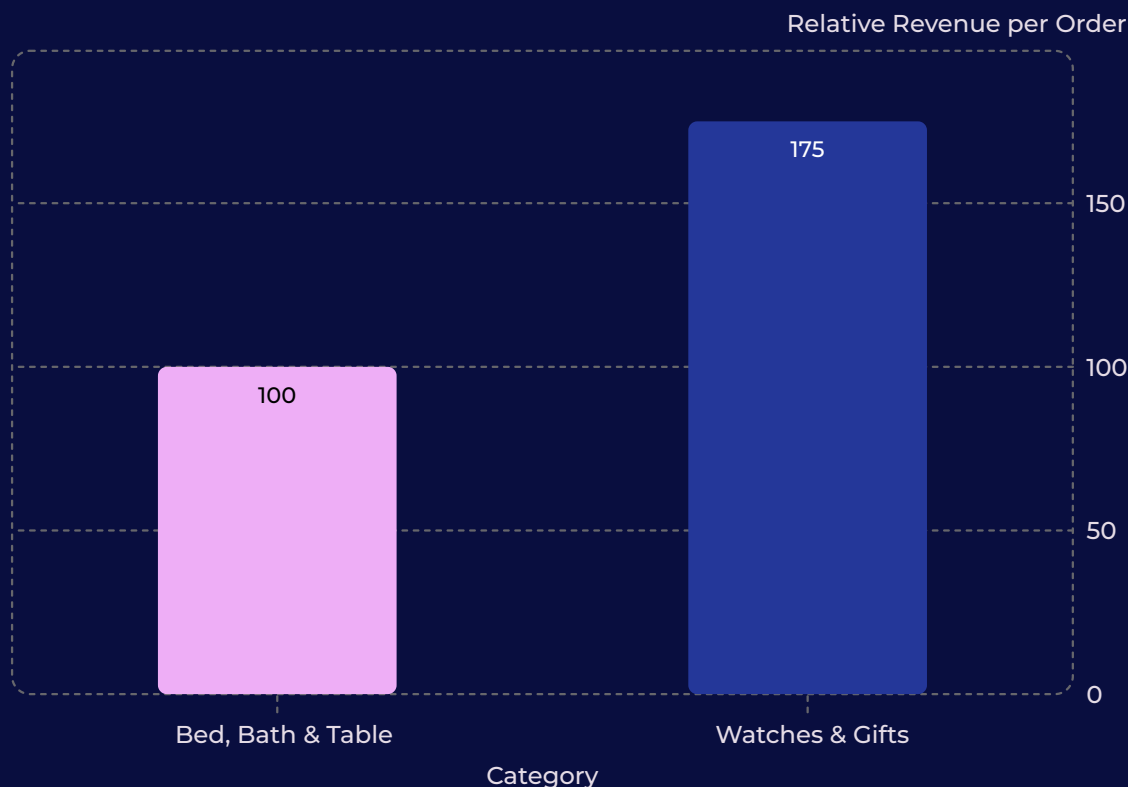
Overall platform average

A Logistics Gap, Not a Systemic Failure

Late deliveries are **concentrated in northern and northeastern states** — regions geographically distant from the São Paulo seller base.

This is a **last-mile infrastructure gap**, not a platform-wide reliability problem. Targeted regional logistics investment is the right lever.

Revenue Per Order Varies Dramatically by Category



Higher Value, Lower Volume

Watches & Gifts generate approximately **75% more revenue per order** than Bed, Bath & Table — despite half the transaction volume.

A blanket pricing or promotional strategy underserves both segments. **Category-specific margin strategies** are essential to unlocking full revenue potential.

Customer Loyalty: RFM Segmentation



Champions

High-frequency, high-value buyers. Prioritize VIP retention programs and early product access.



At Risk

Previously active customers showing declining engagement. Targeted win-back campaigns with personalized offers.

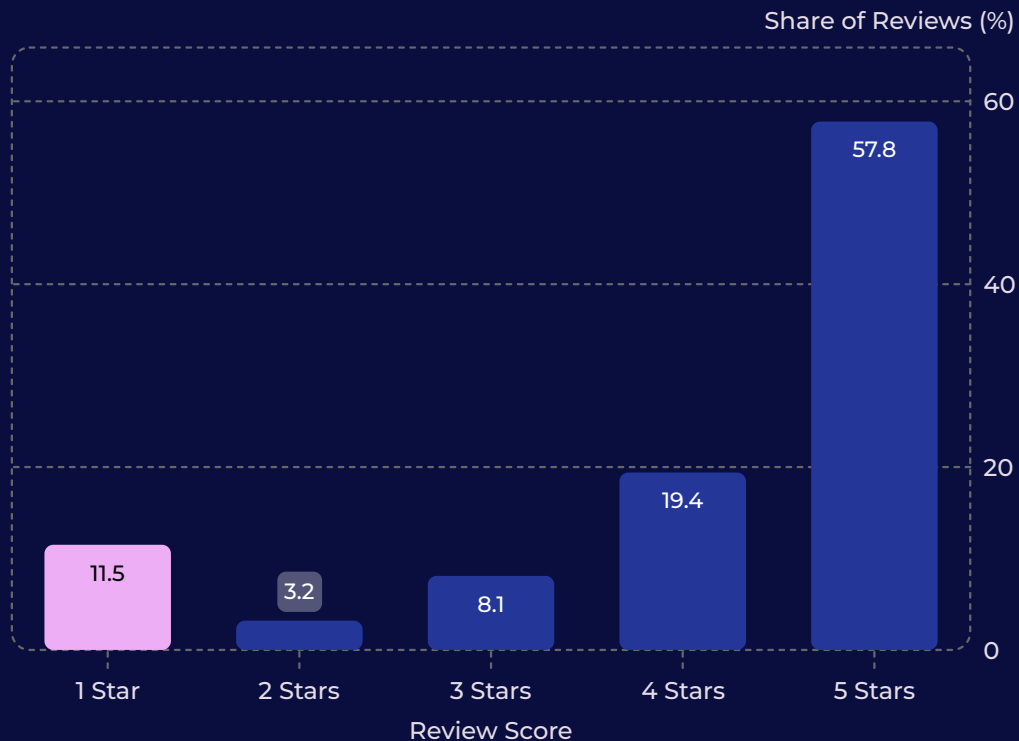


Lost Customers

Long-dormant accounts. Evaluate cost-benefit of reactivation vs. redirecting budget to acquisition.

RFM segmentation transforms a flat customer list into **actionable lifetime value strategy** — enabling precision targeting and smarter marketing spend.

Data Quality Discovery: The Bimodal Review Score



Unexpected Bimodal Distribution

57.78% five-star reviews — yet one-star is the **second most common score**. The middle is hollowed out.

Root cause identified: a backend timestamp bug combined with mobile UI tap behavior was contaminating the delivery-satisfaction signal.

❏ This is not a customer experience anomaly — it's a data integrity issue with real business metric consequences.

The Tech Stack: Enterprise-Grade Pipeline



Data Storage

SQLite with **Python sqlite3** for structured querying — reproducible, version-controlled, and lightweight.



Analysis & EDA

pandas for cleaning and transformation. **matplotlib** and **seaborn** for exploratory visualization.



Dashboarding

Power BI for executive-ready interactive reporting delivered to business stakeholders.



Version Control

Git / GitHub — full audit trail, collaborative-ready, and production-quality repository structure.

A classical painting of a hand interacting with a modern tablet displaying a Power BI dashboard. The hand is rendered in a realistic, detailed style, with visible skin texture and a gold ring on the ring finger. The tablet screen shows a complex dashboard with various charts, including a bar chart, a line chart, a pie chart, and a map. The background of the painting is dark and atmospheric, with a vase of flowers and a book visible on the left.

Interactive Power BI Dashboard

Page 1: Executive Sales Overview

GMV trends, geographic breakdown, category revenue performance — designed for C-suite consumption at a glance.

Page 2: Customer & Operational Insights

RFM segment distribution, delivery performance by region, and review score analysis for operational teams.

5 interactive slicers: Date range · Customer state · Product category · RFM segment · Shipping type — enabling any stakeholder to drill into their specific question without needing a data analyst in the room.

Key Takeaway: Data-Driven Decision Making

Revenue Drivers Identified

Geographic concentration, category margin variance, and seasonality patterns are now visible and actionable — not hidden in raw tables.

Customer Segments Defined

RFM segmentation turns 100,000+ anonymous transactions into **named, targetable customer groups** with distinct retention strategies.

Operational Gaps Pinpointed

Delivery latency mapped to specific geographies — transforming a vague reliability concern into a **targeted logistics investment case**.

Hidden Issues Surfaced

The bimodal review score anomaly turned a technical data quality finding into a **strategic insight about metric validity** — the kind of discovery that changes how leadership interprets satisfaction scores.